PC INTERFACE
USB-B Connector (power in) --> protections --> PWR switch --> 5V rail

J1
USB B Female Connector

F1
2920L050_72DR

D1
SMAJ5.0A

D2
SP0502BAHTG

FL1
BLM21SP601SN1D

S2
L101011MS02Q
SYSTEM POWER SWITCH

POWER RAILS
LDO to 3V3
LDO to 1V2

IC2
MIC5504-3.3YM5-TR

IC3
MIC5504-1.2YM5-TR

FPGA ICE40UP5K
SYSTEM BRAIN

IC1
ICE40UP5K-SG48I

FPGA POWER PINS

FPGA VCCIO_0
FPGA VCCIO_2
FPGA SPI_VCCIO_1
FPGA VCC1
FPGA VCC_2
FPGA VPP_2V5

pin2: 3V3_VREF only — do not drive

J2
61300611121

PROGRAMMING INTERFACE
Connector for programming
SPI config flash memory

Pullup/pulldowns

DO NOT PLACE R68
(by default)

IC5
W25Q40CLSNIG

FPGA CONFIG LOGIC
Reset and Done

S1
OS102011MA1QN1

FTDI (USB-UART bridge IC) VPHY and VPLL filter circuits

FB1
BLM18AG601SN1D

FB2
BLM18AG601SN1D

FTDI (USB-UART bridge IC) oscillator circuit

Y1
830208214809

FTDI POWER PINS

FTDI VCORE_1 decoupling
FTDI VCORE_2 decoupling
FTDI VCORE_3 decoupling
FTDI VCCIO_1 decoupling
FTDI VCCIO_2 decoupling
FTDI VCCIO_3 decoupling
FTDI VCCIO_4 decoupling
FTDI VREGOUT filter cap
FTDI VREGIN decoupling and bulk

FTDI debug LEDs (UART RX and TX)

FPGA DEBUG HEADER
FPGA debug outputs for routing internal signals to external logic analyzer/scope.

J4
61300821121

UART PROBE HEADER
UART lines between FPGA and FTDI (USB-UART bridge IC) for external logic analyzer/scope.

J5
61300821121

CONFIG PROBE HEADER
CONFIG signals from FPGA (CRESET_B AND CDONE) for external logic analyzer/scope.

J6
61300311121

POWER RAIL DEBUG HEADER
External rail injection for debug only.

J7
61300621121

FTDI CHIP
(UART-USB BRIDGE IC)

IC4
FT2232HL-TRAY

Integrated voltage regulator output

INPUT AUXILIARIES
- series resistors
- weak pulldowns
- Schmitt buffers

LA inputs support 3V3 / 5V digital logic only.
No RS232, negative voltage, analog input, or arbitrary overvoltage.
Floating inputs default low through 100kOhm pulldowns.

IC6
SN74LVC14ADBRE4

IC7
SN74LVC14ADBRE4

IC8
SN74LVC14ADBRE4

5A low =< 5Y high
4A low =< 4Y high

SAMPLING CONNECTOR

J3
61202021621

Indicator LEDs

add layout label:
DONE_LED_EN

Q1
2N7002H6327XTSA2

Sheet: /	
File: iCE40UP5K_logic_analyzer_rev2.kicad_sch	
Title: iCE40UP5K Logic Analyzer Rev2	
Size: A3	Date: 19/06/2026
KiCad E.D.A. 9.0.6	Rev: A
	Id: 1/1